

Project Initialization and Planning Phase

Date	12 July 2026
Team ID	XXXXXX
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

OptiCrop is an AI-powered crop recommendation system that helps farmers identify the most suitable crop based on soil nutrients (Nitrogen, Phosphorus, Potassium), temperature, humidity, pH, and rainfall. Using a trained machine learning model, the system provides accurate crop recommendations to improve agricultural productivity, reduce crop failure, and support informed farming decisions.

Project Overview	
Objective	Develop an intelligent crop recommendation system that predicts the most suitable crop using soil and environmental parameters to improve agricultural productivity.
Scope	The system assists farmers by analyzing soil and weather conditions, predicting the best crop, and providing data-driven recommendations through a simple web application.
Problem Statement	
Description	Farmers often face difficulties in selecting the right crop due to changing environmental conditions and lack of accurate decision-support systems, leading to low yield and financial loss.
Impact	An incorrect crop choice can reduce productivity, increase farming costs, and negatively affect farmers' income and food production.
Proposed Solution	
Approach	Use a Machine Learning model trained on agricultural datasets to analyze soil nutrients and climatic conditions, then recommend the most suitable crop through a Flask-based web application.
Key Features	- AI-based crop recommendation. - Predicts crop using N, P, K, temperature, humidity, pH, and rainfall.

	• Displays prediction confidence. • User-friendly web interface. • Fast and accurate recommendations.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/Ryzen 5 or above
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	10 GB free SSD/HDD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, Pandas, NumPy, Joblib
Development Environment	IDE	Jupyter Notebook, VS Code
Data		
Data	Source, size, format	Crop Recommendation Dataset (CSV from Kaggle), 2200 records, CSV format